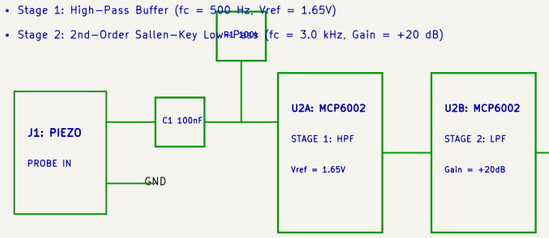
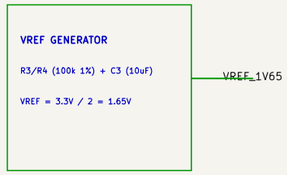
BLOCK 1: ANALOG FRONT-END (AFE) & ACOUSTIC FILTERING

- *Piezo Contact Transducer (27mm PZT Ceramic clamped to pipe)*
- *Dual Rail-to-Rail Op-Amp: Microchip MCP6002-I/SN*
- Stage 1: High-Pass Buffer (fc = 500 Hz, Vref = 1.65V)
- Stage 2: 2nd-Order Sallen-Key Low-Pass (fc = 3.0 kHz, Gain = +20 dB)

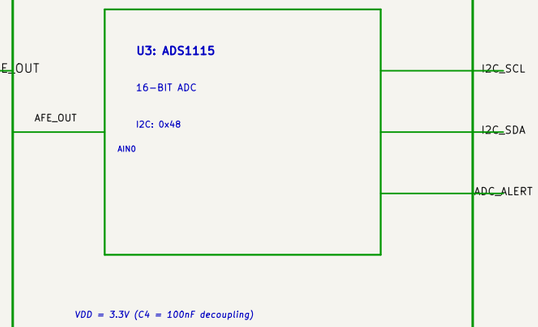


Output: 500Hz-3kHz (±1.5Vpp centered at 1.65V)



BLOCK 2: 16-BIT PRECISION ADC (TI ADS1115)

- *Texas Instruments ADS1115IDGSR (16-Bit Delta-Sigma)*
- Dynamic Range: 16-Bit | Programmable Gain: 1x to 16x
- Sampling Rate: 860 SPS / Fast Burst Mode
- I2C Interface: Address 0x48 (ADDR pin tied to GND)



VDD = 3.3V (C4 = 100nF decoupling)

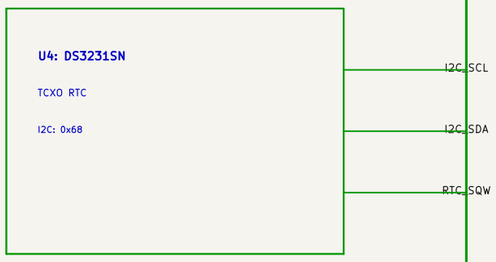
BLOCK 3: ESP32-S3 DUAL-CORE COMPUTE & ESP-NOW MESH

- *Espressif ESP32-S3-WROOM-1 (Xtensa Dual-Core 32-bit LX7 @ 240 MHz)*
- Core 0: High-Speed ADC Sampling, Circular Waveform Buffer (2048 samples)
- Core 1: ESP-NOW Mesh Protocol, TDOA Cross-Correlation, Consensus Engine
- Memory: 8MB Flash + 8MB Octal PSRAM | Radio: 2.4GHz Wi-Fi/BLE (ESP-NOW <20ms)



BLOCK 4: MICROSECOND PRECISION TCXO RTC (DS3231)

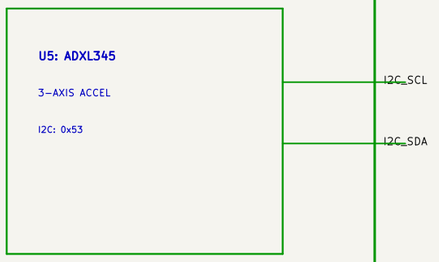
- *Analog Devices / Maxim DS3231SN (Integrated Temp-Compensated TCXO)*
- Frequency Accuracy: ±2 ppm (0°C to +40°C) | Aging: ±1 ppm/year
- Crucial for TDOA cross-correlation microsecond node synchronization
- Backup Supply: Onboard CR1220 Coin Cell / Supercapacitor backup



SQW output gives 1Hz / 32kHz microsecond reference pulse

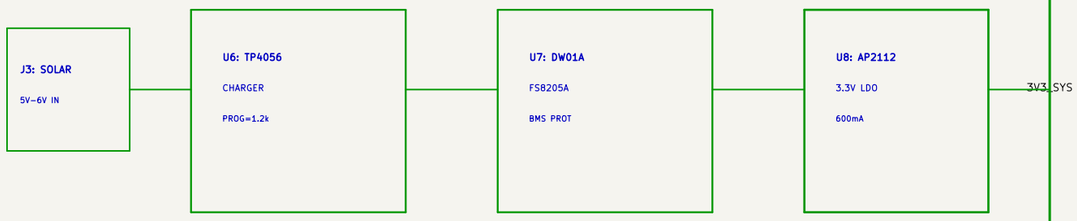
BLOCK 5: ADXL345 AUX ACCELEROMETER

- *Analog Devices ADXL345BCCZ 3-Axis MEMS*
- Verifies mechanical pipe acceleration vs airborne noise
- False-positive rejection: surface rumbles disregarded
- I2C Interface: Address 0x53 (ALT_ADDRESS low)



BLOCK 6: SOLAR BMS, LI-ION PROTECTION & LOW-NOISE REGULATION

- *Solar Input: 5V/6V Monocrystalline Panel (J3) + TP4056 CC/CV Lithium Charger (U6)*
- Battery Cell: 3.7V 3000mAh Grade-A Li-ion 18650 Pack with DW01A + FS8205A Over/Under Protection (U7)
- System Rail: Diodes Inc AP2112K-3.3 ultra-low-noise 600mA LDO (ripple <30uV, PSRR = 65dB)
- Fully battery-isolated & intrinsically safe: Zero mains connection, zero pipe electrolytic hazard



Battery Input: J2 (JST-PH 2.0mm) | C_in = 10uF X7R | C_out = 22uF Low-ESR Ceramic

Team TECHTONICS | Vishwakarma Awards 2026-27 (Host: IIT Hyderabad)

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File: AQUAPULSE_Node.kicad_sch

Title: AQUAPULSE: Non-Invasive Acoustic Sensor Node (ESP32-S3 + AFE + A

Size: A3	Date: 2026-10-25	**Rev: v1.2-RELEASE**
KiCad E.D.A. 10.0.7		Id: 1/1